

Ph.D Thesis Introductory Chapter

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Contents

1 Modern Data and Probabilistic Algorithms	1
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1 Modern Data and Probabilistic Algorithms

As we venture further into the 21st century, our world is shaped, built, or dare we say, defined by data (Hey et al. [2009]). From simple information like class rosters, to large scale datasets like the Human Genome Project, we are recording everything, with the realization that there are limitless values to be extracted. Mathematics and algorithms are at the center of this grand picture, as the extent to which we can unlock this value is fundamentally constrained by our mathematical understanding. Indeed, for decades have we collected visual data such as satellite imagery and medical scans, but the studies and utilization of them remained primarily manual. In 2012, the success of ImageNet(Krizhevsky et al. [2012]), along with Convolution Neural Network (CNN), made automated analysis of visual data possible. It was moments like this that revealed to us there is so much to be seen and discovered with data.

Among all the mathematical theories and tools that forge the foundation of data science, probability theory and probabilistic algorithms are of unique and paramount importance. With data comes uncertainty and noise, and with forecasting and prediction comes ambiguity. The nature of the tasks data science aims to tackle and the intrinsic problems stemming from all forms of data call for probabilistic approaches. It provides the mathematical language and framework for us to navigate through randomness with deterministic confidence.

Take the example of CNN we just mentioned. The success of CNN is seen by many as the watershed moment in the history of mathematics and computer science. Though the advancements in semiconductor industry and hardware engineering should never be overlooked, the birth of CNN is the result of a series of brilliant developments in probability since Kholmogorov. The applications of concepts like softmax layers, maximum log-likelihood estimation, etc all have a deep root in probability theory and constitute the core mechanism of CNN(Krizhevsky et al. [2012]). And our stories have just begun. As of today, the development of Large Language Models (LLM), the closest ever creation of humans to Artificial General Intelligence (AGI) is essentially a probabilistic algorithm (Vaswani et al. [2017]).

In this thesis, we will explore three different probabilistic algorithms. In chapter 2, we will study t-SNE, another famous and important dimension reduction algorithm developed by Hinton (van der Maaten and Hinton [2008]). Will there still be an equilibrium measure for t-SNE if we swap out its input and output kernels? How does the performance differ? Those

are the questions we aim to answer. In chapter 3, we will transform a classic theorem, Lovász Local Lemma, into a probabilistic algorithm that can learn difficult combinatorial structures. We will also provide the theoretical foundation of the algorithm, proving its correctness and giving an estimation of the running time expectation. In chapter 4, we will dive into another dimension reduction algorithm, Variational Autoencoder (VAE), and how a particularly designed VAE can handle missing data via imputation and help us through factor analysis (thus the name VarFA). All three algorithms represent a different yet extremely important aspect of data science, each with its own charm and characteristics.

For the rest of this chapter, we will provide a brief introduction to the probabilistic algorithms we will discuss in this thesis. From basic concepts, to history, and related progress. It is very exciting to see all the mathematical concepts our predecessors have developed, and have been passionate about, are being turned into new technologies that impact and steer the modern world. And may this thesis provides an insignificant, yet meaningful addition to this great painting of our future.

References

- [1] Tony Hey, Stewart Tansley, Kristin Tolle, and Jim Gray. *The Fourth Paradigm: Data-Intensive Scientific Discovery*. Microsoft Research, October 2009. ISBN 978-0-9825442-0-4. URL <https://www.microsoft.com/en-us/research/publication/fourth-paradigm-data-intensive-scientific-discovery/>.
- [2] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. Imagenet classification with deep convolutional neural networks. In F. Pereira, C.J. Burges, L. Bottou, and K.Q. Weinberger, editors, *Advances in Neural Information Processing Systems*, volume 25. Curran Associates, Inc., 2012. URL https://proceedings.neurips.cc/paper_files/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf.
- [3] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf.
- [4] Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. *Journal of Machine Learning Research*, 9(2579-2605), 2008.
- [5] Antonio Auffinger and Daniel Fletcher. Equilibrium distributions for t-distributed stochastic neighbour embedding, 2023, 2304.03727. URL <https://arxiv.org/abs/2304.03727>.
- [6] Aditya Ravuri and Neil D. Lawrence. Towards one model for classical dimensionality reduction: A probabilistic perspective on umap and t-sne, 2024, 2405.17412. URL <https://arxiv.org/abs/2405.17412>.
- [7] Pavlin G. Poličar and Blaž Zupan. Visualizing high-dimensional temporal data using direction-aware t-sne, 2024, 2403.19040. URL <https://arxiv.org/abs/2403.19040>.
- [8] Stéphane Boucheron, Gábor Lugosi, and Pascal Massart. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press, 2013.
- [9] Ben Weinkove. Dimension reduction and the gradient flow of relative entropy, 2024, 2409.16963. URL <https://arxiv.org/abs/2409.16963>.

- [10] Wenjun Liu, Quoc Ngo, and Vu Huy. Several interesting integral inequalities. *Journal of Mathematical Inequalities*, 3, 06 2009.
- [11] Dmitry Kobak, George Linderman, Stefan Steinerberger, Yuval Kluger, and Philipp Berens. Heavy-tailed kernels reveal a finer cluster structure in t-sne visualisations. In Ulf Brefeld, Elisa Fromont, Andreas Hotho, Arno Knobbe, Marloes Maathuis, and Céline Robardet, editors, *Machine Learning and Knowledge Discovery in Databases*, pages 124–139, Cham, 2020. Springer International Publishing. ISBN 978-3-030-46150-8.
- [12] Geoffrey E. Hinton. Training products of experts by minimizing contrastive divergence. *Neural Comput.*, 14(8):1771–1800, 2002.
- [13] Ioannis Tsochantaridis, Thorsten Joachims, Thomas Hofmann, and Yasemin Altun. Large margin methods for structured and interdependent output variables. *J. Mach. Learn. Res.*, 6:1453–1484, December 2005.
- [14] Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron C. Courville, and Yoshua Bengio. Generative adversarial nets. In *NIPS*, pages 2672–2680, 2014.
- [15] Diederik P. Kingma and Max Welling. Auto-encoding variational bayes. In *ICLR*, 2014.
- [16] Mathieu Germain, Karol Gregor, Iain Murray, and Hugo Larochelle. MADE: masked autoencoder for distribution estimation. In *ICML*, volume 37 of *JMLR Workshop and Conference Proceedings*, pages 881–889. JMLR.org, 2015.
- [17] Hugo Larochelle and Iain Murray. The neural autoregressive distribution estimator. In *AISTATS*, volume 15 of *JMLR Proceedings*, pages 29–37. JMLR.org, 2011.
- [18] Aäron van den Oord, Nal Kalchbrenner, and Koray Kavukcuoglu. Pixel recurrent neural networks. In *ICML*, volume 48 of *JMLR Workshop and Conference Proceedings*, pages 1747–1756. JMLR.org, 2016.
- [19] Martín Arjovsky, Soumith Chintala, and Léon Bottou. Wasserstein generative adversarial networks. In *ICML*, volume 70 of *Proceedings of Machine Learning Research*, pages 214–223. PMLR, 2017.
- [20] Yang Song and Stefano Ermon. Generative modeling by estimating gradients of the data distribution. In *NeurIPS*, pages 11895–11907, 2019.
- [21] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *ICLR*. OpenReview.net, 2021.
- [22] Kevin P. Murphy, Yair Weiss, and Michael I. Jordan. Loopy belief propagation for approximate inference: An empirical study. In *UAI*, pages 467–475. Morgan Kaufmann, 1999.
- [23] Jonathan S. Yedidia, William T. Freeman, and Yair Weiss. Generalized belief propagation. In *NIPS*, pages 689–695. MIT Press, 2000.
- [24] Martin J. Wainwright and Michael I. Jordan. Log-determinant relaxation for approximate inference in discrete markov random fields. *IEEE Trans. Signal Process.*, 54(6-1):2099–2109, 2006.
- [25] A. Braunstein, M. Mézard, and R. Zecchina. Survey propagation: an algorithm for satisfiability. *Random Struct. Algorithms*, 27:201–226, 2005.

- [26] Henrik Reif Andersen, Tarik Hadzic, John N. Hooker, and Peter Tiedemann. A constraint store based on multivalued decision diagrams. In *CP*, volume 4741 of *Lecture Notes in Computer Science*, pages 118–132. Springer, 2007.
- [27] Mark Chavira, Adnan Darwiche, and Manfred Jaeger. Compiling relational bayesian networks for exact inference. *Int. J. Approx. Reason.*, 42(1-2):4–20, 2006.
- [28] Pascal Van Hentenryck. *Constraint Satisfaction in Logic Programming*. MIT Press, Cambridge, MA, USA, 1989. ISBN 0-262-08181-4.
- [29] Vibhav Gogate and Rina Dechter. Importance sampling-based estimation over and/or search spaces for graphical models. *Artificial Intelligence*, 184-185:38 – 77, 2012.
- [30] Tian Sang, Paul Bearne, and Henry Kautz. Performing bayesian inference by weighted model counting. In *AAAI*, AAAI’05, pages 475–481, 2005.
- [31] Matt J. Kusner, Brooks Paige, and José Miguel Hernández-Lobato. Grammar variational autoencoder. In *ICML*, volume 70, pages 1945–1954. PMLR, 2017.
- [32] Wengong Jin, Regina Barzilay, and Tommi S. Jaakkola. Junction tree variational autoencoder for molecular graph generation. In *ICML*, volume 80 of *Proceedings of Machine Learning Research*, pages 2328–2337. PMLR, 2018.
- [33] Hanjun Dai, Yingtao Tian, Bo Dai, Steven Skiena, and Le Song. Syntax-directed variational autoencoder for structured data. In *ICLR (Poster)*. OpenReview.net, 2018.
- [34] Zhiting Hu, Zichao Yang, Xiaodan Liang, Ruslan Salakhutdinov, and Eric P. Xing. Toward controlled generation of text. In *ICML*, volume 70 of *Proceedings of Machine Learning Research*, pages 1587–1596. PMLR, 2017.
- [35] Daniel Lowd and Pedro M. Domingos. Learning arithmetic circuits. In *UAI*, pages 383–392. AUAI Press, 2008.
- [36] Fan Ding, Jianzhu Ma, Jinbo Xu, and Yexiang Xue. XOR-CD: linearly convergent constrained structure generation. In *ICML*, volume 139 of *Proceedings of Machine Learning Research*, pages 2728–2738. PMLR, 2021.
- [37] Paul ErdHos and László Lovász. Problems and results on 3-chromatic hypergraphs and some related questions. In *Colloquia Mathematica Societatis Janos Bolyai 10. Infinite and Finite Sets, Keszthely (Hungary)*. Citeseer, 1973.
- [38] Robin A. Moser and Gábor Tardos. A constructive proof of the general lovász local lemma. *J. ACM*, 57(2):11:1–11:15, 2010.
- [39] Heng Guo, Mark Jerrum, and Jingcheng Liu. Uniform sampling through the lovász local lemma. *J. ACM*, 66(3):18:1–18:31, 2019.
- [40] Nan Jiang, Yi Gu, and Yexiang Xue. Learning markov random fields for combinatorial structures with sampling through lovász local lemma, 2022.
- [41] Daniel Selsam, Matthew Lamm, Benedikt Bünz, Percy Liang, Leonardo de Moura, and David L. Dill. Learning a SAT solver from single-bit supervision. In *ICLR (Poster)*. OpenReview.net, 2019.
- [42] Haonan Duan, Pashootan Vaezipoor, Max B. Paulus, Yangjun Ruan, and Chris J. Maddison. Augment with care: Contrastive learning for combinatorial problems. In *ICML*, volume 162 of *Proceedings of Machine Learning Research*, pages 5627–5642. PMLR, 2022.
- [43] Zhuwen Li, Qifeng Chen, and Vladlen Koltun. Combinatorial optimization with graph convolutional networks and guided tree search. In *NeurIPS*, pages 537–546, 2018.

- [44] Emre Yolcu and Barnabás Póczos. Learning local search heuristics for boolean satisfiability. In *NeurIPS*, pages 7990–8001, 2019.
- [45] Xinyun Chen and Yuandong Tian. Learning to perform local rewriting for combinatorial optimization. In *NeurIPS*, pages 6278–6289, 2019.
- [46] Nikolaos Karalias and Andreas Loukas. Erdos goes neural: an unsupervised learning framework for combinatorial optimization on graphs. In *NeurIPS*, pages 6659–6672, 2020.
- [47] Yoshua Bengio, Andrea Lodi, and Antoine Prouvost. Machine learning for combinatorial optimization: A methodological tour d’horizon. *Eur. J. Oper. Res.*, 290(2): 405–421, 2021.
- [48] Jayanta Mandi, Emir Demirovic, Peter J. Stuckey, and Tias Guns. Smart predict-and-optimize for hard combinatorial optimization problems. In *AAAI*, pages 1603–1610. AAAI Press, 2020.
- [49] Radford M Neal. *Probabilistic inference using Markov chain Monte Carlo methods*. Department of Computer Science, University of Toronto Toronto, ON, Canada, 1993.
- [50] Paul Dagum and R. Martin Chavez. Approximating probabilistic inference in bayesian belief networks. *IEEE Trans. Pattern Anal. Mach. Intell.*, 15(3):246–255, 1993.
- [51] Hong Ge, Kai Xu, and Zoubin Ghahramani. Turing: A language for flexible probabilistic inference. In *AISTATS*, volume 84, pages 1682–1690. PMLR, 09–11 Apr 2018.
- [52] Alexander T. Ihler, John W. Fisher III, and Alan S. Willsky. Loopy belief propagation: Convergence and effects of message errors. *J. Mach. Learn. Res.*, 6:905–936, 2005.
- [53] Amin Coja-Oghlan, Noëla Müller, and Jean B. Ravelomanana. Belief propagation on the random k-sat model, 2020.
- [54] Fan Ding and Yexiang Xue. Contrastive divergence learning with chained belief propagation. In *PGM*, volume 138 of *Proceedings of Machine Learning Research*, pages 161–172. PMLR, 2020.
- [55] Vibhav Gogate and Rina Dechter. Samplesearch: Importance sampling in presence of determinism. *Artif. Intell.*, 175(2):694–729, 2011.
- [56] Carla P. Gomes, Ashish Sabharwal, and Bart Selman. Near-uniform sampling of combinatorial spaces using XOR constraints. In *NIPS*, pages 481–488. MIT Press, 2006.
- [57] Stefano Ermon, Carla P. Gomes, Ashish Sabharwal, and Bart Selman. Taming the curse of dimensionality: Discrete integration by hashing and optimization. In *ICML (2)*, volume 28 of *JMLR Workshop and Conference Proceedings*, pages 334–342. JMLR.org, 2013.
- [58] Dimitris Achlioptas and Panos Theodoropoulos. Probabilistic model counting with short xors. In *SAT*, volume 10491 of *Lecture Notes in Computer Science*, pages 3–19. Springer, 2017.
- [59] Supratik Chakraborty, Kuldeep S. Meel, and Moshe Y. Vardi. A scalable and nearly uniform generator of SAT witnesses. In *CAV*, volume 8044, pages 608–623. Springer, 2013.
- [60] Yishay Mansour. Learning boolean functions via the fourier transform. In *Theoretical advances in neural computation and learning*, pages 391–424. Springer, 1994.

- [61] Chris K Carter and Robert Kohn. On gibbs sampling for state space models. *Biometrika*, 81(3):541–553, 1994.
- [62] Rahul Gupta, Shubham Sharma, Subhajit Roy, and Kuldeep S. Meel. WAPS: weighted and projected sampling. In *TACAS*, volume 11427, pages 59–76, 2019.
- [63] Supratik Chakraborty, Daniel J. Fremont, Kuldeep S. Meel, Sanjit A. Seshia, and Moshe Y. Vardi. Distribution-aware sampling and weighted model counting for SAT. In *AAAI*, pages 1722–1730. AAAI Press, 2014.
- [64] Stefano Ermon, Carla P. Gomes, Ashish Sabharwal, and Bart Selman. Embed and project: Discrete sampling with universal hashing. In *NIPS*, pages 2085–2093, 2013.
- [65] Fan Ding and Yexiang Xue. XOR-SGD: provable convex stochastic optimization for decision-making under uncertainty. In *UAI*, volume 161 of *Proceedings of Machine Learning Research*, pages 151–160. AUAI Press, 2021.
- [66] Priyanka Golia, Mate Soos, Sourav Chakraborty, and Kuldeep S. Meel. Designing samplers is easy: The boon of testers. In *FMCAD*, pages 222–230. IEEE, 2021.
- [67] Rafael Dutra, Kevin Laeuffer, Jonathan Bachrach, and Koushik Sen. Efficient sampling of SAT solutions for testing. In *ICSE*, pages 549–559. ACM, 2018.
- [68] Mate Soos, Stephan Gocht, and Kuldeep S. Meel. Tinted, detached, and lazy CNF-XOR solving and its applications to counting and sampling. In *CAV (1)*, volume 12224 of *Lecture Notes in Computer Science*, pages 463–484. Springer, 2020.
- [69] Shubham Sharma, Rahul Gupta, Subhajit Roy, and Kuldeep S. Meel. Knowledge compilation meets uniform sampling. In *LPAR*, volume 57 of *EPiC Series in Computing*, pages 620–636, 2018.
- [70] Carmine Dodaro and Alessandro Previti. Minipref: A tool for preferences in SAT (short paper). In *RCRA + RiCeRcA*, volume 2538 of *CEUR Workshop Proceedings*. CEUR-WS.org, 2019.
- [71] Massimo Lauria, Jan Elffers, Jakob Nordström, and Marc Vinyals. Cnfgn: A generator of crafted benchmarks. In *SAT*, volume 10491, pages 464–473. Springer, 2017.
- [72] Alexey Ignatiev, António Morgado, and João Marques-Silva. Pysat: A python toolkit for prototyping with SAT oracles. In *SAT*, volume 10929 of *Lecture Notes in Computer Science*, pages 428–437. Springer, 2018.
- [73] Sourav Chakraborty and Kuldeep S. Meel. On testing of uniform samplers. In *AAAI*, pages 7777–7784, 2019.
- [74] Henry Cohn, Robin Pemantle, and James Gary Propp. Generating a random sink-free orientation in quadratic time. *Electron. J. Comb.*, 9(1), 2002.
- [75] Junji Takahashi, Takuya Yamaguchi, Kosuke Sekiyama, and Toshio Fukuda. Communication timing control and topology reconfiguration of a sink-free meshed sensor network with mobile robots. *IEEE/ASME transactions on mechatronics*, 14(2):187–197, 2009.
- [76] Mark Jerrum. Fundamentals of partial rejection sampling, 2021.
- [77] Emanuele Di Rosa, Enrico Giunchiglia, and Barry O’Sullivan. Optimal stopping methods for finding high quality solutions to satisfiability problems with preferences. In *SAC*, pages 901–906. ACM, 2011.
- [78] Nicolas Prevot, Mate Soos, and Kuldeep S. Meel. Leveraging gpus for effective clause sharing in parallel SAT solving. In *SAT*, pages 471–487, 2021.

- [79] Muhammad Osama Mahmoud. *GPU Enabled Automated Reasoning*. PhD thesis, Mathematics and Computer Science, March 2022.
- [80] Johannes Klaus Fichte, Markus Hecher, and Markus Zisser. An improved gpu-based SAT model counter. In *CP*, pages 491–509, 2019.
- [81] Hao Cen, Kenneth Koedinger, and Brian Junker. Learning factors analysis – a general method for cognitive model evaluation and improvement. In Mitsuru Ikeda, Kevin D. Ashley, and Tak-Wai Chan, editors, *Proc. Intelligent Tutoring Systems*, pages 164–175, 2006. ISBN 978-3-540-35160-3.
- [82] Chris Piech, Jonathan Bassen, Jonathan Huang, Surya Ganguli, Mehran Sahami, Leonidas J Guibas, and Jascha Sohl-Dickstein. Deep knowledge tracing. In *Proc. Advances in Neural Information Processing Systems*, pages 505–513. Curran Associates, Inc., 2015.
- [83] Andrew S. Lan, Andrew E. Waters, Christoph Studer, and Richard G. Baraniuk. Sparse factor analysis for learning and content analytics. *Journal of Machine Learning Research*, 15:1959–2008, 2014.
- [84] Neil T. Heffernan and Cristina Heffernan. The assistments ecosystem: Building a platform that brings scientists and teachers together for minimally invasive research on human learning and teaching. *International Journal of Artificial Intelligence in Education*, 24:470–497, 2014.
- [85] J. Stamper, A. Niculescu-Mizil, S. Ritter, G.J. Gordon, and K.R. Koedinger. Algebra i 2006-2007. *Development data set from KDD Cup 2010 Educational Data Mining Challenge*, 2010.
- [86] J. Stamper, A. Niculescu-Mizil, S. Ritter, G.J. Gordon, and K.R. Koedinger. Bridge to algebra 2006-2007. *Development data set from KDD Cup 2010 Educational Data Mining Challenge*, 2010.
- [87] John C. Stamper and Zachary A. Pardos. The 2010 kdd cup competition dataset: Engaging the machine learning community in predictive learning analytics. In *Journal of Learning Analytics*, 2016.
- [88] Wim J van der Linden and Ronald K Hambleton. *Handbook of modern item response theory*. Springer Science & Business Media, 2013.
- [89] Terry A Ackerman. Using multidimensional item response theory to understand what items and tests are measuring. *Applied Measurement in Education*, 7(4):255–278, 1994.
- [90] Philip I. Pavlik, Hao Cen, and Kenneth R. Koedinger. Performance factors analysis –a new alternative to knowledge tracing. In *Proc. Artificial Intelligence in Education*, page 531–538, NLD, 2009. IOS Press. ISBN 9781607500285.
- [91] Johan Ludwig William Valdemar Jensen et al. Sur les fonctions convexes et les inégalités entre les valeurs moyennes. *Acta Mathematica*, 30:175–193, 1906.
- [92] Diederik P Kingma, Max Welling, et al. An introduction to variational autoencoders. *Foundations and Trends® in Machine Learning*, 12(4):307–392, 2019.
- [93] Danilo Jimenez Rezende, Shakir Mohamed, and Daan Wierstra. Stochastic backpropagation and approximate inference in deep generative models. In *Proc. International Conference on Machine Learning, ICML’14*, page II–1278–II–1286. JMLR.org, 2014.
- [94] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *Proc. International Conference on Learning Representations*, 2015.

- [95] Tian Qi Chen, Xuechen Li, Roger B Grosse, and David K Duvenaud. Isolating sources of disentanglement in variational autoencoders. In *Proc. Advances in Neural Information Processing Systems*, pages 2610–2620, 2018.
- [96] Hyunjik Kim and Andriy Mnih. Disentangling by factorising. In *Proc. International Conference on Machine Learning*, volume 80, pages 2649–2658, 2018.
- [97] Irina Higgins, Loic Matthey, Arka Pal, Christopher Burgess, Xavier Glorot, Matthew Botvinick, Shakir Mohamed, and Alexander Lerchner. beta-vae: Learning basic visual concepts with a constrained variational framework. In *Proc. International Conference on Learning Representations*, 2017.
- [98] Elliot Creager, David Madras, Joern-Henrik Jacobsen, Marissa Weis, Kevin Swersky, Toniann Pitassi, and Richard Zemel. Flexibly fair representation learning by disentanglement. In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proc. Machine Learning Research*, pages 1436–1445, Long Beach, California, USA, 09–15 Jun 2019. PMLR.
- [99] Emile Mathieu, Tom Rainforth, N Siddharth, and Yee Whye Teh. Disentangling disentanglement in variational autoencoders. In *Proc. International Conference on Machine Learning*, pages 4402–4412, Jun 2019.
- [100] Narayanaswamy Siddharth, Brooks Paige, Jan-Willem Van de Meent, Alban Desmaison, Noah Goodman, Pushmeet Kohli, Frank Wood, and Philip Torr. Learning disentangled representations with semi-supervised deep generative models. In *Proc. Advances in Neural Information Processing Systems*, pages 5925–5935, Dec. 2017.
- [101] Shengjia Zhao, Jiaming Song, and Stefano Ermon. Infovae: Balancing learning and inference in variational autoencoders. In *Proc. AAAI Conference on Artificial Intelligence*, volume 33, pages 5885–5892, Feb. 2019.
- [102] Sabine Graf and Kinshuk. *Personalized Learning Systems*, pages 2594–2596. Springer US, Boston, MA, 2012. ISBN 978-1-4419-1428-6. URL https://doi.org/10.1007/978-1-4419-1428-6_152.
- [103] Valerie J Shute and Joseph Psotka. Intelligent tutoring systems: Past, present, and future., 1994.
- [104] John R Anderson, Albert T Corbett, Kenneth R Koedinger, and Ray Pelletier. Cognitive tutors: Lessons learned. *Journal of the Learning Sciences*, 4(2):167–207, 1995.
- [105] Benjamin D Nye, Arthur C Graesser, and Xiangen Hu. Autotutor and family: A review of 17 years of natural language tutoring. *International Journal of Artificial Intelligence in Education*, 24(4):427–469, 2014.
- [106] Neil T Heffernan and Cristina Lindquist Heffernan. The assistments ecosystem: Building a platform that brings scientists and teachers together for minimally invasive research on human learning and teaching. *International Journal of Artificial Intelligence in Education*, 24(4):470–497, 2014.
- [107] Anna N Rafferty, Michelle M LaMar, and Thomas L Griffiths. Inferring learners’ knowledge from their actions. *Cognitive Science*, 39(3):584–618, 2015.
- [108] Cristóbal Romero and Sebastián Ventura. Educational data mining: a review of the state of the art. *IEEE Transactions on Systems, Man, and Cybernetics*, 40(6):601–618, 2010.
- [109] Ryan SJD Baker and Kalina Yacef. The state of educational data mining in 2009: A review and future visions. *Journal of Educational Data Mining*, 1(1):3–17, 2009.

- [110] Agathe Merceron and Kalina Yacef. Educational data mining: a case study. In *Proc. Artificial Intelligence in Education*, pages 467–474, 2005.
- [111] Benoît Choffin, Fabrice Popineau, Yolaine Bourda, and Jill-Jënn Vie. Das3h: Modeling student learning and forgetting for optimally scheduling distributed practice of skills. In *Proc. Educational Data Mining*, 2019.
- [112] Jill-Jënn Vie and Hisashi Kashima. Knowledge tracing machines: Factorization machines for knowledge tracing. In *Proc. AAAI Conference on Artificial Intelligence*, volume 33, pages 750–757, 2019.
- [113] Robert V Lindsey, Jeffery D Shroyer, Harold Pashler, and Michael C Mozer. Improving students’ long-term knowledge retention through personalized review. *Psychological Science*, 25(3):639–647, 2014.
- [114] Michael C Mozer and Robert V Lindsey. Predicting and improving memory retention: Psychological theory matters in the big data era. In *Big Data in Cognitive Science*, pages 43–73. Psychology Press, 2016.
- [115] Kevin H Wilson, Yan Karklin, Bojian Han, and Chaitanya Ekanadham. Back to the basics: Bayesian extensions of irt outperform neural networks for proficiency estimation. In *Proc. Educational Data Mining*, 2016.
- [116] Jiani Zhang, Xingjian Shi, Irwin King, and Dit-Yan Yeung. Dynamic key-value memory networks for knowledge tracing. In *Proc. International Conference on World Wide Web*, pages 765–774, 2017.
- [117] Sein Minn, Yi Yu, Michel C Desmarais, Feida Zhu, and Jill-Jënn Vie. Deep knowledge tracing and dynamic student classification for knowledge tracing. In *Proc. IEEE International Conference on Data Mining*, pages 1182–1187. IEEE, 2018.
- [118] T. Yang, C. Studer, R. Baker, N. Heffernan, and A. S. Lan. Active learning for student affect detection. In *Proc. Educational Data Mining*, 2019.
- [119] Gregory Aist, Barry Kort, Rob Reilly, Jack Mostow, and Rosalind Picard. Experimentally augmenting an intelligent tutoring system with human-supplied capabilities: adding human-provided emotional scaffolding to an automated reading tutor that listens. In *Proc. IEEE International Conference on Multimodal Interfaces*, pages 483–490. IEEE, 2002.
- [120] Jean-Paul Fox. *Bayesian item response modeling: Theory and applications*. Springer Science & Business Media, 2010.
- [121] Andrew Gelman, John B Carlin, Hal S Stern, David B Dunson, Aki Vehtari, and Donald B Rubin. *Bayesian data analysis*. Chapman and Hall/CRC, 2013.
- [122] Zachary A Pardos, Neil T Heffernan, Brigham Anderson, Cristina L Heffernan, and Worcester Public Schools. Using fine-grained skill models to fit student performance with bayesian networks. *Handbook of Educational Data Mining*, 417, 2010.
- [123] Edgar C Merkle. A comparison of imputation methods for bayesian factor analysis models. *Journal of Educational and Behavioral Statistics*, 36(2):257–276, 2011.
- [124] Hua-Hua Chang, Zhiliang Ying, et al. Nonlinear sequential designs for logistic item response theory models with applications to computerized adaptive tests. *The Annals of Statistics*, 37(3):1466–1488, 2009.
- [125] Duanli Yan, Alina A Von Davier, and Charles Lewis. *Computerized multistage testing: Theory and applications*. CRC Press, 2016.

- [126] Kurt VanLehn. Student modeling. *Foundations of Intelligent Tutoring Systems*, 55: 78, 1988.
- [127] Konstantina Chrysafiadi and Maria Virvou. Student modeling approaches: A literature review for the last decade. *Expert Systems with Applications*, 40(11):4715–4729, 2013.
- [128] Walter R Gilks, Sylvia Richardson, and David Spiegelhalter. *Markov chain Monte Carlo in practice*. Chapman and Hall/CRC, 1995.
- [129] Marcy Perkins Driscoll. *Psychology of learning for instruction*. Allyn & Bacon, 1994.
- [130] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The elements of statistical learning: data mining, inference, and prediction*. Springer Science & Business Media, 2009.
- [131] Jean-Paul Fox and Cees AW Glas. Bayesian estimation of a multilevel irt model using gibbs sampling. *Psychometrika*, 66(2):271–288, 2001.
- [132] Robert J Mislevy. Bayes modal estimation in item response models. *Psychometrika*, 51(2):177–195, 1986.
- [133] Min Chi, Kenneth R Koedinger, Geoffrey J Gordon, Pamela Jordon, and Kurt VanLahn. Instructional factors analysis: A cognitive model for multiple instructional interventions. *Educational Data Mining*, 2011.
- [134] Kikumi K. Tatsuoka. Rule space: An approach for dealing with misconceptions based on item response theory. *Journal of Educational Measurement*, 20(4):345–354, 1983. ISSN 00220655, 17453984. URL <http://www.jstor.org/stable/1434951>.
- [135] Christopher M Bishop. *Pattern recognition and machine learning*. springer, 2006.
- [136] Wenbo Gong, Sebastian Tschiatschek, Richard Turner, Sebastian Nowozin, and José Miguel Hernández-Lobato. Icebreaker: Element-wise active information acquisition with bayesian deep latent gaussian model. In *Proc. Advances in Neural Information Processing Systems*, 2019.
- [137] Chao Ma, Sebastian Tschiatschek, Konstantina Palla, José Miguel Hernández-Lobato, Sebastian Nowozin, and Cheng Zhang. Eddi: Efficient dynamic discovery of high-value information with partial vae. In *Proc. International Conference on Machine Learning*, 2019.
- [138] Chao Ma, Wenbo Gong, José Miguel Hernández-Lobato, Noam Koenigstein, Sebastian Nowozin, and Cheng Zhang. Partial vae for hybrid recommender system. In *Proc. NeurIPS Workshop on Bayesian Deep Learning*, 2018.
- [139] Dawen Liang, Rahul G Krishnan, Matthew D Hoffman, and Tony Jebara. Variational autoencoders for collaborative filtering. In *Proc. International Conference on World Wide Web*, pages 689–698, 2018.
- [140] Ruslan Salakhutdinov, Andriy Mnih, and Geoffrey Hinton. Restricted boltzmann machines for collaborative filtering. In *Proc. International Conference on Machine Learning*, pages 791–798, 2007.
- [141] Suvash Sedhain, Aditya Krishna Menon, Scott Sanner, and Lexing Xie. Autorec: Autoencoders meet collaborative filtering. In *Proc. International Conference on World Wide Web*, pages 111–112, 2015.
- [142] Alfredo Nazabal, Pablo M Olmos, Zoubin Ghahramani, and Isabel Valera. Handling incomplete heterogeneous data using vaes. *arXiv preprint arXiv:1807.03653*, 2018.

- [143] Mike Wu, Richard L Davis, Benjamin W Domingue, Chris Piech, and Noah Goodman. Variational item response theory: Fast, accurate, and expressive. *arXiv preprint arXiv:2002.00276*, 2020.
- [144] Zoubin Ghahramani and Matthew J Beal. Variational inference for bayesian mixtures of factor analysers. In *Proc. Advances in Neural Information Processing Systems*, pages 449–455, 2000.
- [145] Jian-hua Zhao and LH Philip. A note on variational bayesian factor analysis. *Neural Networks*, 22(7):988–997, 2009.
- [146] Martín Abadi et al. TensorFlow: Large-scale machine learning on heterogeneous systems, 2015. URL <https://www.tensorflow.org/>. Software available from [tensorflow.org](https://www.tensorflow.org/).
- [147] Adam Paszke et al. Pytorch: An imperative style, high-performance deep learning library. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, and R. Garnett, editors, *Proc. Advances in Neural Information Processing Systems*, pages 8024–8035. Curran Associates, Inc., 2019.
- [148] David JC MacKay and David JC Mac Kay. *Information theory, inference and learning algorithms*. Cambridge University Press, 2003.
- [149] Hemant Ishwaran, J Sunil Rao, et al. Spike and slab variable selection: frequentist and bayesian strategies. *The Annals of Statistics*, 33(2):730–773, 2005.
- [150] Michalis K Titsias and Miguel Lázaro-Gredilla. Spike and slab variational inference for multi-task and multiple kernel learning. In *Proc. Advances in neural information processing systems*, pages 2339–2347, 2011.
- [151] Francesco Tonolini, Bjorn Sand Jensen, and Roderick Murray-Smith. Variational sparse coding, July 2019. URL <http://eprints.gla.ac.uk/191553/>.
- [152] Danilo Jimenez Rezende and Shakir Mohamed. Variational inference with normalizing flows. In *Proc. International Conference on Machine Learning - Volume, ICML'15*, page 1530–1538. JMLR.org, 2015.
- [153] Durk P Kingma, Tim Salimans, Rafal Jozefowicz, Xi Chen, Ilya Sutskever, and Max Welling. Improved variational inference with inverse autoregressive flow. In *Proc. Advances in Neural Information Processing Systems*, pages 4743–4751, 2016.
- [154] Ruqi Zhang, Chunyuan Li, Jianyi Zhang, Changyou Chen, and Andrew Gordon Wilson. Cyclical stochastic gradient mcmc for bayesian deep learning. *Proc. International Conference on Learning Representations*, 2020.
- [155] Raza Habib and David Barber. Auxiliary variational mcmc. *Proc. International Conference on Learning Representations*, 2019.
- [156] Yi-An Ma, Tianqi Chen, and Emily Fox. A complete recipe for stochastic gradient mcmc. In *Proc. Advances in Neural Information Processing Systems*, pages 2917–2925, 2015.
- [157] Nando De Freitas, Pedro Højen-Sørensen, Michael I Jordan, and Stuart Russell. Variational mcmc. In *Proc. conference on Uncertainty in Artificial Intelligence*, pages 120–127. Morgan Kaufmann Publishers Inc., 2001.
- [158] Jiaming Song, Shengjia Zhao, and Stefano Ermon. A-nice-mc: Adversarial training for mcmc. In *Proc. Advances in Neural Information Processing Systems*, pages 5140–5150, 2017.